

NOURA SHOUSHA

Undergrad AI Engineer

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PROFESSIONAL SUMMARY

AI Engineering student (CGPA 3.63/4.0) specializing in agentic AI, ML systems, and workflow automation. Comfortable across the full pipeline — from data prep and model training to deployment — using Python, TensorFlow, and Scikit-learn, and have shipped several retrieval-augmented generation (RAG) systems built on Qdrant. Also work with n8n for automating AI workflows. Currently gaining hands-on experience through an ML engineering internship at FlyRank and a generative/agentic AI traineeship with DEPI, alongside a track record of hackathon builds and side projects. Looking for ML/AI engineering and AI automation roles where I can help turn working prototypes into reliable, production-ready systems.

TECHNICAL SKILLS

- Programming Languages: Python, SQL, C++, Go
- ML / AI: Machine Learning, Recommendation Systems, Agentic AI, TensorFlow, Scikit-learn, Pandas, NLP, Transformers
- Automation & Web: n8n (workflow automation), HTML/CSS
- Tools & Platforms: Git, GitHub, Docker, Socket Programming, Concurrency (Go)
- Other: IoT Basics, Data Structures & Algorithms

PROFESSIONAL EXPERIENCE

- Forward Program Participant** — *McKinsey.org* Sep 2026 – Present
- Participating in a 10-week professional development program covering career and workplace skills.
- AI Trainee — DEPI Round 5 (Generative & Agentic AI)** — *Digital Egypt Pioneers Initiative* Jul 2026 – Present
- Training in generative and agentic AI system development as part of DEPI's Round 5.
- ML Engineering Intern** — *FlyRank (Online Internship Program)* Jul 2026 – Present
- Working on applied machine learning tasks as part of a remote engineering team.
 - Building hands-on experience with production-style ML workflows.
- AI Engineering Intern** — *BWAI-MASR Edition* Apr 2026 – May 2026
- Studied cloud computing and agentic AI concepts through hands-on training.
 - Explored intelligent automation systems and AI deployment workflows.
- ML Engineering Intern** — *DecodeLabs* May 2026 – Jun 2026
- Built end-to-end machine learning projects, from data preparation to model deployment.
- AI/ML Engineering Intern** — *CodeAlpha* May 2026 – Jun 2026
- Delivered AI-focused technical tasks, strengthening applied model implementation skills.
- Part-Time AI System Manager** — *Tattbiqat – Ana Al-Katib Application* Jul 2025 – Sep 2025
- Managed the text-to-speech AI system during pre-deployment of a reading and writing community platform for the Arab region.
 - Coordinated with the engineering team to ensure AI system reliability ahead of launch.

HIGHLIGHTED PROJECTS

MedSeek – Pediatric & Neonatal Health RAG Assistant — github.com/MedInfant-RAG/MedSeek-Infant-Health-Assistant

- Multilingual (100+ languages) RAG system for pediatric and neonatal health, deployed as a Telegram bot with voice and image input support; built dual response personas for parents (first-aid and emergency guidance) versus doctors (detailed clinical answers). Built for the Orange | Instant AI Hackathon.
- Tech: Python, RAG, Qdrant, BGE-m3 Embeddings, OpenAI GPT-4o-mini, Telegram Bot API

Brain Tumor Prediction System — github.com/mo-elgazar74/Brain-Tumor-Classification-and-Care

- ML model to classify brain tumors from medical data with high accuracy using TensorFlow and Scikit-learn.
- Tech: Python, TensorFlow, Scikit-learn, Pandas

ReadNext – Book Recommendation System — github.com/nrshousha/ReadNext

- Content-based book recommendation engine that suggests titles based on user reading history and preferences.
- Tech: Python, Scikit-learn, Pandas, ML

Egypt Law RAG – Arabic Legal Q&A System — github.com/the-ever-shining-Manal/egypt-law-rag

- Dockerized retrieval-augmented generation system answering questions on the Egyptian Penal Code, using a Qdrant vector database; built and integrated the core AI pipeline end-to-end and owned automation and orchestration on a multi-person team.
- Tech: Python, Docker, Qdrant, RAG, NLP

Inventory Intelligence – Text-to-SQL Inventory Query Engine — github.com/nrshousha/text-to-sql

- Built a natural-language-to-SQL pipeline over a PostgreSQL inventory database, using OpenAI function calling for structured SQL generation; designed a multi-layer safety architecture (unconditional DDL blocking, human-in-the-loop confirmation for write queries, prompt-injection mitigation) to safely support read and write queries generated from natural language.
- Tech: Python, FastAPI, PostgreSQL, Docker, OpenAI API, SQL

Emotion Detection System — github.com/nrshousha/emotion-analysis

- Text-based emotion classification model built by fine-tuning a DistilRoBERTa transformer.
- Tech: Python, Transformers, DistilRoBERTa, NLP

Traffic Light Optimization System — github.com/the-ever-shining-Manal/Traffic-Light-Optimization-Team

- Simulated traffic light control using multiple algorithms (BFS, greedy, priority queues) to minimize vehicle wait time.
- Tech: Python, Algorithms, Data Structures

Titanic Survival Prediction — github.com/nrshousha/ML-project-Titanic-Survival

- Predictive classification model for passenger survival using feature engineering and ensemble methods.
- Tech: Python, Scikit-learn, Pandas, ML

EDUCATION

B.Sc. in Artificial Intelligence Engineering — *Tanta University, Egypt*

Aug 2022 – Expected Jun 2027

- CGPA: 3.63 / 4.0

CERTIFICATIONS

- BWAI-MASR Edition – Google & ITI
- AI Hackathon – Orange & Instant & Creativa
- Machine Learning, data science and AI engineering with python – Udemy

LANGUAGES

- Arabic (Native) | English (Professional)

EXTRACURRICULAR ACTIVITIES

- Vice President of Student Activities — led organizing teams for university events.
- Participated in technical presentations and collaborative AI engineering projects.